

Fluid Mechanics: Lecture Notes

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We Finished the chapter about elasticity and now starting the fluid mechanics chapter of the course.

1 Introduction to Fluid Kinematics

Fluid kinematics is the study of how fluids flow and how to describe fluid motion without considering the forces that cause the motion.

- A solid in rest - stable equilibrium (long term order).
- Fluid in rest - Neutral equilibrium (שייט)
- נוזלים וגזים. Liquids and gasses are in different phase there are differences and similarities.

$$\rho = \rho_0 \quad \text{Incompressible fluid}$$

Consider a water tank in rest. We have internal forces of the kind:

$$dF_i = -pd\vec{S} \quad (1)$$

where $p = p(\vec{x})$ is the pressure. Note: we have only isotropic part and the deviatoric part is vanish.

$$\tau_{ij} = -p\delta_{ij} \quad (2)$$

If theoretically we had deviatoric part it would cause a flow.

$$dF_i = \sigma_{ij}dS_j = -p\delta_{ij}dS_j = -pdS_i \Rightarrow d\vec{F} = -pd\vec{S} \quad (3)$$

2 Units

$$\text{Pressure} = \frac{\text{Force}}{L^2}$$

$$1 \text{ Pascal} = 1 \frac{N}{m^2} \setminus \text{m.k.s}$$

$$1 \text{ Atm} = 1 \frac{\text{Kg} \cdot \text{F}}{\text{cm}^2} = \frac{9.8 \text{ Newton}}{10^{-4} m^2} = 1.01 \times 10^5 \text{ Pascal} \quad (4)$$

$$760 \text{ mm Hg} = 1 \text{ Atm} \quad 1 \text{ Bar} = 10^5 \text{ Pa}$$

$$\text{Psi} = 1 \frac{\text{pound} \times F}{\text{inch}^2} = \frac{0.45 \text{ Kg} \times F}{(2.54)^2 \times \text{cm}^2} = 0.07 \text{ Atm}$$

3 Hydrostatics

Stable equilibrium when $\frac{d^2V}{dx^2}|_{x_0} > 0$ The hydrostatic equation are including the internal force and the external force.

$$\partial_{ij}\sigma_{ij} + \underbrace{\rho g_i}_{f_{\text{ext}}} = 0 \quad (5)$$

Substitute Equation 1

$$\begin{aligned} \partial_{ij}\sigma_{ij} + \underbrace{\rho g_i}_{f_{\text{ext}}} &= -\partial_j(p\delta_{ij}) + \rho g_i \\ &= -\partial_i p + \rho g_i \end{aligned} \quad (6)$$

$$0 = -\vec{\nabla}p + \rho\vec{g} \quad (7)$$

case 1

Incompressible fluid:

$$\rho = \rho_0 \quad (8)$$

Solution:

$$0 = \rho gh \quad (9)$$

where h is the depth coordinate. we can choose h as z so we would get:

$$\vec{\nabla}p = \rho g \hat{z} \Rightarrow p = \rho gz \quad (10)$$

Problem - The pressure difference at water depth

Remark - the pressure is changing at rate of 1 Atm every 10 meters, Let's derive it

$$\frac{p}{h} = \rho g \quad (11)$$

$$\rho_{\text{water}} = 1 \frac{\text{gr}}{\text{cm}^3} = 10^3 \frac{\text{Kg}}{\text{m}^3} \text{ Hence,}$$

$$\frac{p}{h} = \rho g = 10^3 \times 9.8 \simeq 10^4 \frac{\text{Pa}}{\text{m}} = \frac{10^5 \text{ Pa}}{10\text{m}} \quad (12)$$

corollary: Water tank with $h \ll \frac{p}{\rho g}$ For this tank the pressure is constant

$$\nabla p = 0 \Rightarrow p = \text{Const.} \quad (13)$$

This result called Pascal's Principle. This principle has usages in engineering and form the basis for hydraulics. Oil has density of $\sim 0.9 \frac{\text{g}}{\text{cm}^3}$

Case 2 - Compresable fluid

For a gas, the density is not fixed. Air near the ground is denser than air high in the atmosphere. To determine $p(z)$ we need, in addition to force balance, a relation between pressure, density, and temperature: an equation of state.

For an ideal gas

$$pV = nRT = Nk_B T \quad (14)$$

Where n is the number of moles, R is the gas constant, N is the number of molecules, k_B is Boltzmann's constant, and T is the absolute temprature measured in kelvin.

In equilibrium, the temprature is constant $T = \text{Const.}$ and therfore there are no heat flow.

$$-\frac{1}{\rho(p)}\vec{\nabla}p = \vec{g} \Rightarrow \vec{\nabla} \int^p = \frac{dg^T}{dp}\vec{\nabla}p = \frac{1}{\rho}\vec{\nabla}p \quad (15)$$

Therefore the hydrostatic equation becomes

$$\vec{\nabla}g_T = \mathbf{g} \quad (16)$$

The function $g_{T(p)}$ packages the compressibility of the fluid into one scalar potential. For incompressible water it reduces to $\frac{p}{\rho_0}$; for a gas it becomes a logarithmic function of pressure.

Connection to Thermodynamics

$$dE = TdS - pdV \quad (17)$$

Transform legnder

$$\left.\frac{\partial G}{\partial p}\right|_T = V. \quad (18)$$

4 Isothermal Atmosphere $T = T_0$

The atmosphere is the canonical compressible hydrostatic example. Unlike water, air has a density that decreases strongly with height. We simplify the real atmosphere by assuming constant temperature; this is not exact, but it captures the central exponential scale.

Let z point upward. Then $\mathbf{g} = -g\hat{z}$ and the hydrostatic equation becomes

$$\frac{dp}{dz} = -\rho g \quad (19)$$

We cannot do a simple integration because ρ is z dependant.

$$p = \frac{\rho}{m}k_B T, \quad \rho \quad (20)$$